KERALA POLYTECHNIC · REVISION 2026

Course 2001A | Semester II | Theory · 4 Credits | 3:1:0:0

Engineering Mathematics – I

Complex Numbers · Probability & Statistics · Integral Calculus · Numerical Methods & Vector Algebra

Complete handbook for the Kerala Diploma Revision 2026 curriculum. Covers all four modules with deep explanations, worked examples, interactive calculators, Malayalam support, and exam-oriented practice.

PROGRAMME	Common to all Engineering Diploma programmes
COURSE TYPE	Theory
CONTACT HOURS	60 (L:45 + T:15)
SELF-LEARNING	60 hours (suggested)
CIE / ESE	40 / 60

MODULE I	Complex Numbers (14 h)
MODULE II	Probability & Statistics (15 h)
MODULE III	Integral Calculus (15 h)
MODULE IV	Numerical Methods & Vector Algebra (16 h)

📄 Official Source Declaration

Source: Kerala Polytechnic Revision 2026 Syllabus — Course 2001A (Engineering Mathematics – I). This handbook is an educational study aid developed for students. It is not an official publication of the examining body.

⚠️ **Verified Inconsistencies:** The syllabus lists Module IV self-learning activities with a typo — "simpson's \(\frac{1}{2}\) rd rule" — which is corrected to **Simpson's 1/3rd rule** throughout this handbook, matching the main syllabus entry. The total self-learning hours sum to 60 as stated. All other data reconcile.

📊 Course Dashboard

60
CONTACT HOURS

4
CREDITS

40+60
CIE / ESE

4
MODULES

📌 **Prerequisites**
Fundamentals of Engineering Mathematics I (Course 1002) — Trigonometry, Determinants, Differentiation.

Teaching & Assessment Scheme

Teaching Scheme	Self-Learning/Week	Total Hours	Credits	CIE	ESE
3:1:0:0 (L:T:P:R)	4 h	60	4	40	60

Module-wise Instructional Hours

Module	Title	Lecture (L)	Tutorial (T)	Total
I	Complex Numbers	10	4	14
II	Probability & Statistics	12	3	15
III	Integral Calculus	11	4	15
IV	Numerical Methods & Vector Algebra	12	4	16
Total			**60**	

🧭 How to Use This Handbook

1
Understand
Read each module's explanation, formulae, and worked examples.

2
Visualise
Study the SVG diagrams and interactive calculators.

3
Apply
Solve the module-wise questions and try the model paper.

4
Revise
Use the Quick Revision cards and glossary before exams.

🎯 Course Outcomes

CO1
Apply concepts of complex numbers to solve problems. (Bloom: Apply)

CO2
Apply concepts of Probability & Statistics in various mathematical problems. (Bloom: Apply)

CO3
Use integration techniques and differential equations to solve basic engineering problems. (Bloom: Apply)

CO4
Apply basic numerical methods and vector algebra techniques to solve simple engineering problems. (Bloom: Apply)

CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

Legends: 1–Low; 2–Medium; 3–High; – No Correlation

📋 Curriculum Coverage Map

Module → Topics → Hours → CO → Handbook Anchor

Module	Major Topics	Hours	CO	Section
I	Complex number system, operations, Argand plane, modulus, amplitude, polar form, problems	14	CO1	Module 1
II	Sets, probability axioms, coins/dice/cards, statistics, frequency distribution, mean/median/mode	15	CO2	Module 2
III	Integration as reverse, standard integrals, substitution, parts, definite integrals, differential equations	15	CO3	Module 3
IV	Interpolation, Lagrange, Simpson's 1/3rd, scalars/vectors, dot/cross product, magnitude, unit vector	16	CO4	Module 4

🗺️ How the Modules Connect

Module 1

Complex Numbers
Foundation for AC circuit analysis, signal processing, and control systems.

Module 2

Probability & Stats
Data analysis, quality control, risk assessment, and decision-making.

Module 3

Integral Calculus
Area, volume, work, and modelling dynamic systems via differential equations.

Module 4

Numerical Methods & Vectors
Approximation, numerical integration, and spatial/physical vector analysis.

MODULE I

Complex Numbers

14 hours (L:10 + T:4) | CO1 | Bloom: Apply

🎯 Learning Goals

- Understand the complex number system
- Perform fundamental operations
- Represent complex numbers graphically
- Find modulus, amplitude, and polar form
- Solve engineering problems

🔑 Key Facts

- $i = \sqrt{(-1)}$
- $z = a + ib$
- $|z| = \sqrt{(a^2 + b^2)}$
- $\arg(z) = \tan^{-1}(b/a)$
- Polar: $z = r(\cos\theta + i\sin\theta)$

The Complex Number System

A **complex number** is a number of the form **z = a + ib**, where **a** and **b** are real numbers, and **i** is the imaginary unit defined by **$i^2 = -1$**.

- **Real part:** Re(z) = a
- **Imaginary part:** Im(z) = b

$$z = a + ib \;\cdot\; a,b \in \mathbb{R} \;\cdot\; i^2 = -1$$

Example: z = 3 + 4i has Re = 3 and Im = 4.

> **Malayalam support:** സങ്കീർണ്ണ സംഖ്യകൾ (complex numbers) എന്നത് a + ib രൂപത്തിലുള്ള സംഖ്യകളാണ്. ഇവിടെ 'i' എന്നത് √(−1) ആണ്. ഇത് ഇലക്ട്രിക്കൽ എഞ്ചിനീയറിംഗിലും സിഗ്നൽ പ്രോസസ്സിംഗിലും വ്യാപകമായി ഉപയോഗിക്കുന്നു.

> 💡 **Tip:** The real and imaginary parts are always real numbers. The imaginary part includes the coefficient of i, not i itself.

Conjugate and Equality

Conjugate: The conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

$$\bar{z} = a - ib$$

Equality: Two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal iff $a_1 = a_2$ **and** $b_1 = b_2$.

Example: If $2x + 3iy = 4 + 6i$, then $2x = 4 \Rightarrow x = 2$, and $3y = 6 \Rightarrow y = 2$.

💡 **Tip:** The conjugate is used to rationalise denominators and find modulus.

Fundamental Operations

For $z_1 = a + ib$ and $z_2 = c + id$:

- **Addition:** $z_1 + z_2 = (a+c) + i(b+d)$
- **Subtraction:** $z_1 - z_2 = (a-c) + i(b-d)$
- **Multiplication:** $z_1 \cdot z_2 = (ac - bd) + i(ad + bc)$
- **Division:** $z_1 / z_2 = (a+ib)/(c+id) \times (c-id)/(c-id) = [(ac+bd) + i(bc-ad)] / (c^2 + d^2)$

Example (Multiplication): $(2+3i)(1-4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$

Example (Division): $(3+4i)/(1-2i) = (3+4i)(1+2i)/5 = (3+6i+4i+8i^2)/5 = (-5+10i)/5 = -1 + 2i$

⚠️ **Common mistake:** In division, always multiply numerator and denominator by the conjugate of the denominator.

Graphical Representation — Argand Plane

Complex numbers are represented on the **Argand plane** (complex plane) with the real part on the horizontal axis and the imaginary part on the vertical axis.

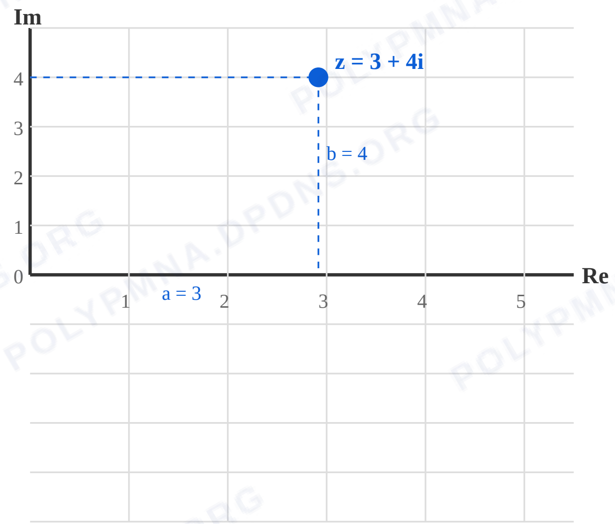

Figure 1: Argand plane showing $z = 3 + 4i$ with real part $a = 3$ and imaginary part $b = 4$.

> **Malayalam support:** Argand plane-ൽ complex numbers-നെ points ആയി represent ചെയ്യുന്നു. Horizontal axis real part-um vertical axis imaginary part-um ആണ്.

Modulus and Amplitude

Modulus (Absolute value): $|z| = \sqrt{a^2 + b^2}$. It is the distance from the origin to the point (a, b).

Amplitude (Argument): $\theta = \arg(z) = \tan^{-1}(b/a)$, measured in degrees (or radians).

$$|z| = \sqrt{a^2 + b^2} \quad \cdot \quad \theta = \tan^{-1}(b/a)$$

Example: For $z = 3 + 4i$: $|z| = \sqrt{9+16} = 5$, $\theta = \tan^{-1}(4/3) \approx 53.13°$

> ⚠️ **Common mistake:** The argument depends on the quadrant. Always check the signs of a and b. In QII, $\theta = 180° - \tan^{-1}(b/|a|)$.

Polar Form

A complex number can be expressed in **polar form** as:

$$z = r(\cos\theta + i\sin\theta) \quad \cdot \quad r = |z|,\ \theta = \arg(z)$$

This is also written as **$z = r\angle\theta$** or **$z = r\,\mathrm{cis}\,\theta$**.

Example: For $z = 3 + 4i$: $r = 5$, $\theta = 53.13°$ ⇒ $z = 5(\cos 53.13° + i \sin 53.13°)$

💡 **Tip:** Polar form is especially useful for multiplication and division — multiply moduli and add arguments.

Solved Problems

Problem 1: Find the modulus and argument of $z = -1 + i$.
Solution: $a = -1$, $b = 1$. $r = \sqrt{(1+1)} = \sqrt{2}$. $\theta = 180° - \tan^{-1}(1/1) = 180° - 45° = 135°$.

Problem 2: Express $z = 2 - 2i$ in polar form.
Solution: $r = \sqrt{(4+4)} = 2\sqrt{2}$. $\theta = -\tan^{-1}(2/2) = -45°$. $z = 2\sqrt{2}(\cos(-45°) + i\sin(-45°))$.

Problem 3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.
Solution: $(1+2i)(3-i) = 3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$.

💡 **Engineering Application:** Complex numbers model AC circuits — impedance $Z = R + iX$, with magnitude giving total opposition and phase angle giving phase shift.

🧮 Complex Number Calculator

Enter real and imaginary parts to compute modulus, argument, and polar form.

Real part (a)

3

Imaginary part (b)

4

z = 3 + 4i | |z| = 5.000 | arg = 53.13° | Polar: 5.000(cos 53.13° + i sin 53.13°)

MODULE II

Probability & Statistics

15 hours (L:12 + T:3) | CO2 | Bloom: Apply

🎯 Learning Goals

- Understand sets, random experiments, sample space
- Apply classical probability axioms
- Solve problems with coins, dice, cards
- Organise data and compute mean/median/mode

🔑 Key Facts

- $P(E) = n(E)/n(S)$
- $0 \leq P(E) \leq 1$
- Mean = $\Sigma x/n$
- Median = middle value (ordered)
- Mode = most frequent value

Sets: Empty, Universal, Subsets

- **Set:** A well-defined collection of objects.
- **Empty set (∅):** A set with no elements.
- **Universal set (U):** The set containing all elements under consideration.
- **Subset:** $A \subseteq B$ if every element of A is also in B.

> **Example:** `For rolling a die, U = {1,2,3,4,5,6}, A = {2,4,6} (even numbers) is a subset of U.`

> **Malayalam support:** Set എന്നത് നിർവചിക്കപ്പെട്ട വസ്തുക്കളുടെ ശേഖരമാണ്. Empty set-ന് ഒരു ഘടകവുമില്ല. Universal set-ൽ എല്ലാ ഘടകങ്ങളും ഉൾപ്പെടും.

Random Experiments, Outcomes, Sample Space

- **Random experiment:** An experiment whose outcome is not certain.
- **Outcome:** A single result of an experiment.
- **Sample space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space.

> **Example:** `Tossing a coin: S = {H, T}. Event E = {H} (getting heads).`

Classical Definition & Axioms of Probability

Classical definition: For a random experiment with n(S) equally likely outcomes, the probability of event E is:

$$P(E) = n(E) / n(S)$$

Axioms:

- $0 \le P(E) \le 1$
- $P(S) = 1$
- For mutually exclusive events, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

> 💡 **Tip:** The sum of probabilities of all outcomes in S is 1.

Problems: Coins, Die, Cards

Coin (1 coin): `S = {H, T}. P(H) = 1/2.`

Coin (2 coins): `S = {HH, HT, TH, TT}. P(HT) = 1/4, P(at least one H) = 3/4.`

Die: `S = {1,2,3,4,5,6}. P(even) = 3/6 = 1/2.`

Cards: `From a deck of 52, P(heart) = 13/52 = 1/4.`

⚠️ **Common mistake:** In "at least one" problems, use the complement rule: P(at least one) = 1 − P(none).

Introduction to Statistics

Statistics: The science of collecting, organising, analysing, and interpreting data.

- **Ungrouped data:** Raw data as collected.
- **Frequency distribution:** A table showing how often each value occurs.

Example: `Marks: 5,7,7,8,9. Frequency: 5→1, 7→2, 8→1, 9→1.`

Measures of Central Tendency

Mean ($\bar{x}$): Sum of all values ÷ number of values.

$$\bar{x} = \Sigma x / n$$

Median: The middle value when data is ordered (or average of two middle values if n is even).

Mode: The value that occurs most frequently.

Example: Data: 3, 5, 7, 7, 9. Mean = 31/5 = 6.2. Median = 7. Mode = 7.

Example (even n): 2, 4, 6, 8. Median = (4+6)/2 = 5.

💡 **Tip:** Mean is affected by outliers; median and mode are more robust.

Frequency Distribution — Mean, Median, Mode

For grouped frequency distribution:

$$\bar{x} = \Sigma(f \cdot x) / \Sigma f$$

Median for grouped data: find the class containing (n+1)/2-th observation.

Mode for grouped data: the class with highest frequency.

Example: x: 1,2,3,4 f: 2,3,1,2. Σf = 8. Σ(f·x) = 2+6+3+8 = 19. $\bar{x}$ = 19/8 = 2.375.

Malayalam support: Frequency distribution-ൽ data-യെ groups ആക്കി തിരിച്ച് അവയുടെ frequencies കണക്കാക്കുന്നു. Mean, median, mode എന്നിവ central tendency-യുടെ measures ആണ്.

🧮 Mean, Median, Mode Calculator

Enter numbers separated by commas (e.g. 5,7,7,8,9).

Data

5,7,7,8,9

Mean: 7.200 | Median: 7.000 | Mode: 7

MODULE III

Integral Calculus

15 hours (L:11 + T:4) | CO3 | Bloom: Apply

🎯 Learning Goals

- Understand integration as reverse of differentiation
- Recall standard integrals
- Apply substitution and integration by parts
- Evaluate definite integrals
- Solve first-order differential equations

🔑 Key Facts

- $\int x^n\, dx = x^{n+1}/(n+1) + C$ $(n \neq -1)$
- $\int e^x\, dx = e^x + C$
- $\int 1/x\, dx = \ln|x| + C$
- $\int \sin x\, dx = -\cos x + C$
- $\int \cos x\, dx = \sin x + C$

Integration as Reverse of Differentiation

Integration is the inverse process of differentiation. If $d/dx\,[F(x)] = f(x)$, then $\int f(x)\,dx = F(x) + C$, where C is the constant of integration.

$$\int f(x)\,dx = F(x) + C \quad \cdot \quad d/dx\,[F(x)] = f(x)$$

Example: $d/dx\,(x^2) = 2x \Rightarrow \int 2x\,dx = x^2 + C$.

💡 **Tip:** The constant C accounts for all antiderivatives. Always include it in indefinite integrals.

Standard Integrals

$\int x^n \, dx = x^{n+1}/(n+1) + C$

$n \neq -1$

$\int 1/x \, dx = \ln|x| + C$

$x \neq 0$

$\int e^x \, dx = e^x + C$

$\int \sin x \, dx = -\cos x + C$

$\int \cos x \, dx = \sin x + C$

$\int \sec^2 x \, dx = \tan x + C$

$\int \mathrm{cosec}^2 x \, dx = -\cot x + C$

$\int \sec x \tan x \, dx = \sec x + C$

$\int \mathrm{cosec}\, x \cot x \, dx = -\mathrm{cosec}\, x + C$

Rules of Integration & Substitution

Rules:

- $\int k \cdot f(x)\,dx = k \cdot \int f(x)\,dx$
- $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

Substitution:

- $\int f(ax+b)\,dx = (1/a)\,F(ax+b) + C$
- $\int f(x) \cdot f'(x)\,dx = [f(x)]^2/2 + C$
- $\int f'(x)/f(x)\,dx = \ln|f(x)| + C$

Example (substitution): $\int (2x+3)^5\,dx$. Let $u = 2x+3$, $du = 2dx \Rightarrow dx = du/2$. $= 1/2 \int u^5\,du = u^6/12 + C = (2x+3)^6/12 + C$.

Example (f'/f): $\int (2x)/(x^2+1)\,dx = \ln|x^2+1| + C$.

Integration by Parts

For product of two functions:

> $\int u\,dv = uv - \int v\,du$

LIATE rule (choose u as the function that comes first): Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

Example: $\int x \sin x\,dx$. Let $u = x$ (algebraic), $dv = \sin x\,dx \Rightarrow du = dx$, $v = -\cos x$. $= x(-\cos x) - \int (-\cos x)\,dx = -x \cos x + \sin x + C$.

> ⚠️ **Common mistake:** Choosing the wrong u can make the integral more complex. Use LIATE.

Definite Integrals

A **definite integral** has limits of integration:

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

Example: $\int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3 - 0 = 1/3$.

Example (area): Area bounded by $y = x^2$, x-axis, $x=0$, $x=2$ is $\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$ sq. units.

💡 **Engineering Application:** Definite integrals compute areas, volumes, work done, and centre of mass.

Differential Equations

Order: The highest derivative present.

Degree: The exponent of the highest derivative (after removing radicals).

Example: $d^2y/dx^2 + 3(dy/dx) + 2y = 0$ → Order 2, Degree 1.

Variable separable: $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x)\,dx$.

Example: $dy/dx = xy \Rightarrow dy/y = x\,dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = C\,e^{(x^2/2)}$.

Linear first-order: $dy/dx + P(x)y = Q(x)$.

$$\text{Integrating factor: I.F.} = e^{(\int P\,dx)} \quad \cdot \quad y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$$

Example: $dy/dx + 2y = x$. $P = 2$, I.F. $= e^{(\int 2\,dx)} = e^{(2x)}$. $y\,e^{(2x)} = \int x\,e^{(2x)}\,dx + C$. Solve to get $y = (2x-1)/4 + C\,e^{(-2x)}$.

🧮 Definite Integral Calculator (polynomial)

Compute $\int_a^b x^n\,dx$. Enter a, b, and n.

a (lower limit)

0

b (upper limit)

2

n (exponent)

2

$\int_a^b x^n\,dx = 2.666667$ (a=0, b=2, n=2)

MODULE IV

Numerical Methods & Vector Algebra

16 hours (L:12 + T:4) | CO4 | Bloom: Apply

🎯 Learning Goals

- Understand interpolation and Lagrange's method
- Apply Simpson's 1/3rd rule
- Distinguish scalars and vectors
- Compute dot and cross products
- Find magnitude and unit vectors

🔑 Key Facts

- Lagrange interpolation: $L(x) = \Sigma\, y_i \cdot \ell_i(x)$
- Simpson's 1/3rd: $\int_a^b f(x)dx \approx h/3\,[f_0 + f_n + 4\Sigma odd + 2\Sigma even]$
- $a \cdot b = |a||b| \cos\theta$
- $a \times b = |a||b| \sin\theta \cdot \hat{n}$

Interpolation & Lagrange's Method

Interpolation is the process of estimating unknown values between known data points.

Lagrange's method constructs a polynomial that passes through all given points:

$$L(x) = \Sigma_{i=0}^{n} y_i \cdot \ell_i(x), \text{ where } \ell_i(x) = \Pi_{j=0,j\neq i}^{n} (x - x_j)/(x_i - x_j)$$

Example: Given (1,2), (2,3), (3,5). Find y at x=2.5.
$\ell_0 = (x-2)(x-3)/((1-2)(1-3)) = (x-2)(x-3)/2$
$\ell_1 = (x-1)(x-3)/((2-1)(2-3)) = -(x-1)(x-3)$
$\ell_2 = (x-1)(x-2)/((3-1)(3-2)) = (x-1)(x-2)/2$
$L(2.5) = 2\cdot\ell_0 + 3\cdot\ell_1 + 5\cdot\ell_2$. Compute: $\ell_0 = (0.5)(-0.5)/2 = -0.125$, $\ell_1 = -(1.5)(-0.5) = 0.75$, $\ell_2 = (1.5)(0.5)/2 = 0.375$. $L = 2(-0.125)+3(0.75)+5(0.375) = -0.25+2.25+1.875 = 3.875$.

💡 **Tip:** Lagrange's method avoids solving systems of equations but can be computationally intensive for large n.

Numerical Integration — Simpson's 1/3rd Rule

For approximating $\int_a^b f(x)\,dx$ with n even subintervals of width $h = (b-a)/n$:

$$\int_a^b f(x)\,dx \approx (h/3)\,[f_0 + f_n + 4(f_1 + f_3 + \ldots + f_{n-1}) + 2(f_2 + f_4 + \ldots + f_{n-2})]$$

Example: Approximate $\int_0^2 x^2\,dx$ using n=4. h = 0.5. $f_0=0$, $f_1=0.25$, $f_2=1$, $f_3=2.25$, $f_4=4$. $\approx (0.5/3)[0+4 + 4(0.25+2.25) + 2(1)] = (1/6)[4 + 4(2.5) + 2] = (1/6)[4+10+2] = 16/6 = 2.6667$. Exact: $8/3 = 2.6667$.

⚠️ **Common mistake:** n must be even for Simpson's 1/3rd rule. If n is odd, use Simpson's 3/8th rule or composite methods.

Scalars and Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, temperature).
- **Vector:** A quantity with both magnitude and direction (e.g., velocity, force).

Unit vectors: $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$ along x, y, z axes.

Position vector: $r = x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}$.

> **Malayalam support:** Scalar-ന് magnitude മാത്രമേ ഉള്ളൂ (ഉദാ: mass). Vector-ന് magnitude-ഉം direction-ഉം ഉണ്ട് (ഉദാ: velocity). Unit vectors î, ĵ, k̂ axes-നെ സൂചിപ്പിക്കുന്നു.

Magnitude and Unit Vector

Magnitude: $|v| = \sqrt{x^2 + y^2 + z^2}$ for $v = x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}$.

Unit vector: $\hat{v} = v / |v|$ (direction of v).

Example: `v = 3î + 4ĵ. |v| = 5. v̂ = (3/5)î + (4/5)ĵ.`

Vector Addition, Subtraction, Scalar Multiplication

- **Addition:** $v + w = (v_1+w_1)\hat{\imath} + (v_2+w_2)\hat{\jmath} + (v_3+w_3)\hat{k}$
- **Subtraction:** $v - w = (v_1-w_1)\hat{\imath} + (v_2-w_2)\hat{\jmath} + (v_3-w_3)\hat{k}$
- **Scalar multiplication:** $c \cdot v = (c\,v_1)\hat{\imath} + (c\,v_2)\hat{\jmath} + (c\,v_3)\hat{k}$

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v+w = 3î+1ĵ, v−w = 1î+5ĵ, 3v = 6î+9ĵ.`

Dot Product

$$v\cdot w = |v||w| \cos\theta \quad \cdot \quad v\cdot w = v_1w_1 + v_2w_2 + v_3w_3$$

- **Angle:** $\cos\theta = (v\cdot w)/(|v||w|)$
- **Perpendicular:** $v \perp w \Leftrightarrow v\cdot w = 0$

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v·w = 2(1)+3(−2) = 2−6 = −4. |v|=√13, |w|=√5, cos θ = −4/(√13·√5) ≈ −0.496.`

Cross Product

$$v\times w = |v||w| \sin\theta \cdot \hat{n} \quad \cdot \quad v\times w = (v_2w_3-v_3w_2)\hat{\imath} - (v_1w_3-v_3w_1)\hat{\jmath} + (v_1w_2-v_2w_1)\hat{k}$$

- **Parallel:** $v \parallel w \Leftrightarrow v\times w = 0$

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v×w = (3·0−0·(−2))î − (2·0−0·1)ĵ + (2·(−2)−3·1)k̂ = −7k̂.`

💡 **Engineering Application:** Cross product gives torque ($\tau = r\times F$) and is used in magnetism and rotational dynamics.

🧮 **Vector Calculator**

Enter components of two vectors (a, b) and compute dot product, cross product, magnitudes, and angle.

a_1	a_2
2	3

a_3	b_1
0	1

b_2	b_3
-2	0

a·b = -4.000 | a×b = (0.000, 0.000, -7.000) | |a| = 3.606 | |b| = 2.236 | θ = 119.74°

📐 Formula Bank

$z = a + ib,\ i^2 = -1$

Complex number

$|z| = \sqrt{(a^2 + b^2)}$

Modulus

$\theta = \tan^{-1}(b/a)$

Argument

$z = r(\cos\theta + i\sin\theta)$

Polar form

$P(E) = n(E)/n(S)$

Probability

$\bar{x} = \Sigma x / n$

Mean

$\int x^n \, dx = x^{n+1}/(n+1) + C$
Power rule

$\int u \, dv = uv - \int v \, du$
Integration by parts

$\int_a^b f(x) \, dx = F(b) - F(a)$
Definite integral

I.F. = $e^{(\int P \, dx)}$
Linear D.E.

$v \cdot w = v_1 w_1 + v_2 w_2 + v_3 w_3$
Dot product

$v \times w = (v_2 w_3 - v_3 w_2)\hat{\imath} - \ldots$
Cross product

🖼️ Diagram Library for Rapid Revision

📐 Argand Plane
Module 1 — Argand Plane showing complex number representation.

📊 Probability Tree
Visualise outcomes for coin/die experiments via tree diagrams (conceptual).

📈 Vector Geometry
Visualise dot and cross products in 3D space. Module 4

📝 Self-Learning Activities (Official)

Week 1-4 · 16 h

Module I — Complex Numbers

- Find conjugate, modulus, amplitude, polar form
- Problems on equality, addition, subtraction
- Multiply and divide complex numbers
- Applications in engineering areas

Week 5-8 · 15 h

Module II — Probability & Statistics

- Problems with coins, die, cards
- Find mean, median, mode
- Frequency table problems
- Applications in engineering

Week 9-12 · 14 h

Module III — Integral Calculus

- Find integrals of functions
- Evaluate definite integrals
- Solve first-order differential equations
- Applications in engineering

Week 13-15 · 15 h

Module IV — Numerical Methods & Vectors

- Interpolation and Simpson's 1/3rd rule problems
- Vector addition, subtraction, scalar multiplication
- Dot product and cross product
- Area and work done problems

Total self-learning hours: 60 (as per syllabus)

📊 Assessment Methodology

CIE / ESE Breakup

Mode	Attendance	CA1	CA2	CA3	CA4	Series Exam (2 modules)	Series Exam (2 modules)	ESE
Duration	—	2h	2h	2h	2h	2h	2h	2.5h
Exam mark	—	15	15	15	15	15	15	60
Converted to	5	15				20		60
Total	40 (CIE) + 60 (ESE)							

Series Examination Pattern

Duration: 2 hours
Total marks: 50 (converted to 20)

- Part A: 2 × 1 mark
- Part B: 2 × 3 marks
- Part C: 2 × 7 marks (with choice)

End Semester Examination

Duration: 2.5 hours
Total marks: 60

- Part A: 1 mark × 12 questions (2 per module)
- Part B: 3 marks × 6 questions (2 out of 3 per module)
- Part C: 7 marks × 6 questions (1 set with choice per module)

❓ Module-wise Questions with Answers

Module I — Complex Numbers

Q1: Find the modulus and argument of $z = 1 + i$.▼

$|z| = \sqrt{1^2+1^2} = \sqrt{2}$. $\theta = \tan^{-1}(1/1) = 45°$.

Q2: Express $z = 2 - 2i$ in polar form.▼

$r = \sqrt{4+4} = 2\sqrt{2}$. $\theta = -45°$. $z = 2\sqrt{2}(\cos(-45°) + i\sin(-45°))$.

Q3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.▼

$(1+2i)(3-i) = 3-i+6i-2i^2 = 3+5i+2 = 5+5i$.

Q4: What is the conjugate of $3 - 4i$?▼

Conjugate is $3 + 4i$.

Module II — Probability & Statistics

Q1: A coin is tossed twice. Find the probability of getting exactly one head.▼

S = {HH, HT, TH, TT}. n(S)=4. E = {HT, TH}. n(E)=2. P = 2/4 = 1/2.

Q2: Find the mean of 3, 5, 7, 7, 9.▼

Mean = (3+5+7+7+9)/5 = 31/5 = 6.2.

Q3: What is the median of 2, 4, 6, 8?▼

n=4 (even). Median = (4+6)/2 = 5.

Q4: A die is rolled. Find P(prime number).▼

S = {1,2,3,4,5,6}. Primes = {2,3,5}. P = 3/6 = 1/2.

Module III — Integral Calculus

Q1: Evaluate $\int 3x^2\,dx$.▼

$\int 3x^2\,dx = 3\cdot(x^3/3) + C = x^3 + C$.

Q2: Evaluate $\int_0^1 x\,dx$.▼

$[x^2/2]_0^1 = 1/2 - 0 = 1/2$.

Q3: Solve $dy/dx = 2x$.▼

$dy = 2x\,dx \Rightarrow \int dy = \int 2x\,dx \Rightarrow y = x^2 + C$.

Q4: Find the order and degree of $d^2y/dx^2 + 3(dy/dx) + 2y = 0$.▼

Order = 2 (highest derivative), Degree = 1 (exponent of highest derivative).

Module IV — Numerical Methods & Vector Algebra

Q1: Given (1,2), (2,3), (3,5), find y at x=2.5 using Lagrange's method.▼

$L(2.5) = 2\cdot\ell_0 + 3\cdot\ell_1 + 5\cdot\ell_2 = 3.875$ (as worked in section).

Q2: Approximate $\int_0^1 x^2\,dx$ using Simpson's 1/3rd with n=2.▼

$h = 0.5$. $f_0=0$, $f_1=0.25$, $f_2=1$. $\approx (0.5/3)[0+1+4(0.25)] = (1/6)[1+1] = 1/3 = 0.3333$.

Q3: Find the magnitude of v = 3î + 4ĵ.▼

$|v| = \sqrt{9+16} = 5$.

Q4: Compute the dot product of a = (2,3) and b = (1,−2).▼

$a\cdot b = 2(1)+3(-2) = 2-6 = -4$.

📄 Practice Model Paper

Note: This is a practice paper based on the official pattern, not an official SITTTR model question paper.

Course 2001A — Engineering Mathematics – I

Duration: 2.5 hours · **Total marks:** 60

All modules are covered. Answer all parts.

Part A (1 mark each — 12 questions, 2 per module)

- **Q1.** What is the conjugate of $2 + 5i$? (M1)
- **Q2.** Find $|3 - 4i|$. (M1)
- **Q3.** A coin is tossed once. What is P(H)? (M2)
- **Q4.** Define sample space. (M2)
- **Q5.** $\int 2x\,dx$ = ? (M3)
- **Q6.** What is the order of $d^2y/dx^2 + y = 0$? (M3)
- **Q7.** What is a unit vector? (M4)
- **Q8.** When are two vectors perpendicular? (M4)
- **Q9-12.** (Additional questions covering all modules)

Part B (3 marks each — 6 questions, 2 out of 3 per module)

- **M1:** Express $1 - i$ in polar form. *OR* Find modulus and argument of $-1 + i$.
- **M2:** Two coins are tossed. Find P(at least one head). *OR* Find mean/median/mode of 2,3,3,5,6.

- **M3:** Evaluate $\int x \sin x \, dx$. *OR* Solve $dy/dx = 2xy$.
- **M4:** Given (1,2), (2,4), (3,6), find y at x=2.5 using Lagrange. *OR* Find $a \cdot b$ and $a \times b$ for a=(1,2,0), b=(0,1,1).

Part C (7 marks each — 6 questions, 1 set with choice per module)

- **M1:** If $z_1 = 2+3i$ and $z_2 = 1-2i$, find z_1+z_2, z_1-z_2, $z_1 \cdot z_2$, and z_1/z_2. *OR* Express $z = 2+2i$ in polar form and find z^3.
- **M2:** A card is drawn from a deck. Find P(heart), P(ace), P(heart or ace). *OR* Given frequency distribution, compute mean, median, mode.
- **M3:** Evaluate $\int_0^2 (x^2+2x)\, dx$. Solve $dy/dx + 2y = e^x$. *OR* Find area under $y = x^2$ from x=0 to x=2.
- **M4:** Use Simpson's 1/3rd to approximate $\int_0^2 x^3 \, dx$ with n=4. Find $a \times b$ for a=(2,1,0), b=(0,3,4). *OR* Lagrange interpolation problem with 4 points.

⚡ Quick Revision

Module I — Complex Numbers

- $z = a + ib$, $i^2 = -1$
- $|z| = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$
- Polar: $z = r(\cos\theta + i \sin\theta)$
- Operations: add real/imag separately

Module II — Probability & Stats

- $P(E) = n(E)/n(S)$
- Mean = $\Sigma x/n$, Median = middle, Mode = most frequent
- Sample space for coins/dice/cards

Module III — Integral Calculus

- $\int x^n \, dx = x^{n+1}/(n+1) + C$
- $\int u \, dv = uv - \int v \, du$
- Definite: $\int_a^b = F(b)-F(a)$
- D.E.: variable separable & linear

Module IV — Num Methods & Vectors

- Lagrange: $L(x) = \Sigma y_i \cdot \ell_i(x)$
- Simpson's: $h/3 [f_0+f_n+4\Sigma odd+2\Sigma even]$
- $a \cdot b = |a||b| \cos\theta$
- $a \times b = |a||b| \sin\theta \cdot \hat{n}$

⚠️ Common Mistakes — Exam Checklist

- ✔️ In complex division, multiply by conjugate of denominator
- ✔️ In probability, check if order matters (permutations vs combinations)
- ✔️ In integration, always add +C for indefinite integrals
- ✔️ In Simpson's rule, n must be even
- ✔️ In vectors, cross product gives a vector, dot product gives a scalar
- ✔️ In differential equations, identify order and degree correctly

📖 Glossary

Term	Meaning
Complex number	$z = a + ib$, where $i^2 = -1$
Modulus	$\lvert z\rvert = \sqrt{(a^2+b^2)}$, distance from origin
Argument	$\theta = \tan^{-1}(b/a)$, angle with real axis
Polar form	$z = r(\cos\theta + i \sin\theta)$
Sample space	Set of all possible outcomes
Probability	$P(E) = n(E)/n(S)$
Mean	Average = $\Sigma x/n$
Median	Middle value of ordered data
Mode	Most frequent value
Integration	Reverse process of differentiation
Definite integral	$\int_a^b f(x)\,dx = F(b) - F(a)$
Differential equation	Equation involving derivatives
Interpolation	Estimating unknown values from known data
Lagrange's method	Polynomial interpolation through given points
Simpson's 1/3rd rule	Numerical integration using parabolic arcs
Scalar	Quantity with only magnitude
Vector	Quantity with magnitude and direction
Dot product	Scalar product: $a\cdot b = \lvert a\rvert\lvert b\rvert \cos\theta$
Cross product	Vector product: $a\times b = \lvert a\rvert\lvert b\rvert \sin\theta\cdot\hat{n}$

📚 Learning Resources

Textbooks

Sl. No.	Title	Author / Publication
1	Mathematics for Industrial Engineering	SITTTR Kalamasser

References

Sl. No.	Title	Author / Publication
1	Higher Engineering Mathematics	B. S. Grewal, Khanna Publishers
2	Higher Engineering Mathematics	B. V. Ramana, Tata McGraw-Hill
3	A text book of Engineering Mathematics	N. P. Bali & Manish Goyal, Laxmi Publications
4	Schau's outlines of Discrete Mathematics (3rd Ed.)	Seymour Lipschutz & Marc Lipson, McGraw Hill
5	Schau's outlines of Probability (3rd Ed.)	Seymour Lipschutz & Marc Lipson, McGraw Hill
6	Schau's outlines of Statistics (6th Ed.)	Dr. Murray R. Spiegel & Larry J. Stephens, McGraw Hill

Web resources: None officially listed in the syllabus.

🔍 Search this Handbook